

OC IT Projects — Chapter 2

OSI Model — Quiz

Answer first, then check the answer key.

1. OSI stands for:

- A) Open Software Integration
- B) Open System Interconnection
- C) Operating System Interface
- D) Online System Integration

2. How many layers are in the OSI Model?

- A) 4
- B) 5
- C) 7
- D) 8

3. Which layer is responsible for physical transmission of bits?

- A) Data Link
- B) Network
- C) Physical
- D) Transport

4. Switches, Bridges, and NICs are examples of:

- A) Layer 1
- B) Layer 2
- C) Layer 3
- D) Layer 4

5. MAC and LLC belong to:

- A) Physical
- B) Data Link
- C) Network
- D) Transport

6. Logical Addressing and Routing are associated with:

- A) Layer 1
- B) Layer 2
- C) Layer 3
- D) Layer 5

7. Which protocol is mentioned under Layer 3?

- A) TCP
- B) UDP
- C) ICMP
- D) FTP

8. TCP and UDP operate at:

- A) Layer 2
- B) Layer 3
- C) Layer 4

D) Layer 7

9. Which layer handles Data Formatting and Encryption?

A) Session

B) Presentation

C) Transport

D) Application

10. Correct sequence from Layer 7 → Layer 1?

A) Application → Presentation → Session → Transport → Network → Data Link → Physical

B) Physical → Data Link → Network → Transport → Session → Presentation → Application

C) Application → Session → Presentation → Transport → Data Link → Network → Physical

D) Application → Transport → Session → Presentation → Network → Data Link → Physical

11. Which layer establishes and maintains sessions?

A) Physical

B) Data Link

C) Session

D) Presentation

12. Which two protocols are covered in the Transport Layer section?

A) HTTP and HTTPS

B) TCP and UDP

C) ICMP and IP

D) FTP and DNS

13. Which layer includes MAC and LLC?

A) Layer 1

B) Layer 2

C) Layer 3

D) Layer 4

14. Route discovery and selection belong primarily to:

A) Physical

B) Data Link

C) Network

D) Presentation

15. NICs, Bridges, and Switches are examples of:

A) Physical

B) Data Link

C) Network

D) Transport

16. Which layer is concerned with data formatting?

A) Session

B) Presentation

C) Network

D) Physical

17. Which best describes Layer 1?

A) Routing and logical addressing

B) TCP and UDP

- C) Bits and physical transmission/media
- D) Data formatting and encryption

18. Which best describes Layer 3?

- A) Physical cabling only
- B) Logical addressing and routing
- C) Session establishment only
- D) Data formatting only

19. Scenario: A system needs to establish and maintain a communication session. Which layer?

- A) Layer 2
- B) Layer 3
- C) Layer 5
- D) Layer 7

20. Scenario: Data formatting and encryption are required. Which layer?

- A) Layer 2
- B) Layer 4
- C) Layer 6
- D) Layer 7

Answer Key

1-B | 2-C | 3-C | 4-B | 5-B

6-C | 7-C | 8-C | 9-B | 10-A

11-C | 12-B | 13-B | 14-C | 15-B

16-B | 17-C | 18-B | 19-C | 20-C

Quick OSI Revision

- L7 Application
- L6 Presentation → Data Formatting + Encryption
- L5 Session → Establish/Maintain Sessions
- L4 Transport → TCP / UDP
- L3 Network → Logical Addressing + Routing + ICMP
- L2 Data Link → MAC + LLC; NICs, Bridges, Switches
- L1 Physical → Bits, physical media, cabling
- Mnemonic: All People Seems To Need Data Processing

First mini-quiz result: 10/10.